NeuroSpark — Deep-Research Findings

Full verified evidence record (proof). Adversarially fact-checked.

Method: 5 search angles → 24 primary sources fetched → 116 falsifiable claims extracted → 25 verified by 3-vote adversarial check (need 2/3 to refute). Result: **24 confirmed, 1 killed**. 106 agent calls.

Question under study: viability of Project NeuroSpark — Stage 1 (done: synthetic-connectome SNN + MSE-based active-inference grid agent), planned Stage 1.5 (real FlyWire connectome, true variational/expected free energy, RL baseline, ablations + statistics, paper), and planned Stage 2 (wetware MEA control). Generated 22 June 2026.

Headline summary. Feasible and technically sound, but novelty must be framed narrowly. A usable ~1000-neuron FlyWire subgraph (directed synapse-count weights + predicted NT signs) is freely extractable (Zenodo v783, Codex, CAVEclient). The broad 'connectome-constrained embodied agent' is established prior art (Lappalainen 2024; NeuroMechFly v2 2024; FlyGM 2026) and active inference on grids is established (FEPS 2025); the defensible niche is the CONJUNCTION — active inference / EFE on a real connectome vs RL — which no located work occupies. The current EFE is MSE-based (not true variational free energy), so the 'active inference' claim is not yet earned. A Stage 1.5 software study is workshop-publishable if scoped honestly; Stage 2 wetware is far higher risk for a student without a wet lab.

Verified Findings (24 confirmed claims, merged to 9 themes)

1. Q1 NOVELTY: The broad 'connectome-constrained embodied agent for fruit-fly control' idea is already established prior art, so NeuroSpark cannot claim to be the first connectome-based embodied agent or connectome model.

confidence: HIGH | vote: 3-0 (merged from claims 0,2,3,4,5)

Evidence. Three independent top-tier lines of prior art establish this. (a) FlyGM (arXiv:2602.17997, 2026) 'directly instantiates the whole-brain connectome of an adult Drosophila as a graph-structured neural controller' for embodied MuJoCo locomotion, and validates the real connectome against degree-preserving-rewired, random, and MLP baselines, showing higher sample efficiency. (b) Lappalainen et al. (Nature 634:1132-1140, 2024) built a connectome-constrained, task-optimized deep mechanistic network from 64 optic-lobe cell types and showed predictions 'agreed with experimental measurements of neural activity across 26 studies.' (c) NeuroMechFly v2 (Nature Methods 2024, s41592-024-02497-y) couples a connectome-constrained visual network to a closed-loop embodied fly-following behavior. Therefore the project must frame novelty narrowly (active-inference control + RL comparison), NOT as connectome modeling per se.

Sources: arxiv.org/abs/2602.17997 · nature.com/articles/s41586-024-07939-3 · biorxiv.org/content/10.1101/2023.09.18.556649

2. Q1 NOVELTY (defensible niche): The SPECIFIC framing — active inference / expected free energy on a real connectome, compared head-to-head against RL — is NOT pre-empted by the located prior art, because all those connectome controllers use reinforcement learning, not active inference.

confidence: HIGH | vote: 3-0 (merged from claims 1,6)

Evidence. FlyGM trains its connectome controller via 'deep reinforcement learning' (imitation init + RL fine-tune) with no mention of active inference, EFE/VFE, or epistemic value. NeuroMechFly v2 explicitly trains its closed-loop controller 'using reinforcement learning' on a POMDP/Gym interface, with no free-energy method. Thus the AI-vs-RL-on-a-real-connectome comparison is genuinely unaddressed. CAVEAT: this novelty is narrow. FlyGM already demonstrates the generic result that real-connectome topology provides a structural inductive bias (better sample efficiency) versus rewired/random/MLP baselines — so NeuroSpark's novelty must lean specifically on the active-inference/FEP angle, not on 'first connectome-constrained embodied controller.'

Sources: arxiv.org/abs/2602.17997 · biorxiv.org/content/10.1101/2023.09.18.556649

3. Q1 NOVELTY (active inference is also not new): Active inference for grid-world navigation, and the 'active inference vs RL' comparison itself, are established research areas — not novel to NeuroSpark.

confidence: HIGH | vote: 3-0 (merged from claims 7,8)

Evidence. FEPS (PLOS One 2025, PMC12410762) derives a policy 'by minimizing the expected free energy' and is tested on 'a navigation task in a partially observable grid.' The same paper notes 'Various reinforcement learning (RL) models performing active inference have been proposed,' and the AI-vs-RL framing is further populated by Friston et al. 2009 'Reinforcement learning or active inference?', 'Active Inference: Demystified and Compared' (Neural Computation 2021), and Millidge 2019. So neither AIF-on-grid nor AIF-vs-RL is individually novel; only their conjunction with a real connectome is. The 'first systematic AI-vs-RL on a real connectome' framing is therefore defensible ONLY if scoped to exactly that conjunction and the literature search is presented honestly.

Sources: ncbi.nlm.nih.gov/pmc/articles/PMC12410762

4. Q2 FLYWIRE DATA REALITY: A usable ~1000-neuron FlyWire subgraph with directed synapse-count weights and predicted neurotransmitter signs is freely and reliably accessible in 2026, via three independent routes (Zenodo bulk download, Codex, CAVEclient).

confidence: HIGH | vote: 3-0 (merged from claims 11,13,14,16,20,21,22,23)

Evidence. Route 1 (bulk, no API): Zenodo record 10676866 (v783, CC-BY-4.0) gives proofread_connections_783.feather (852MB) with directed pre/post root IDs, syn_count weights, and NT averages (gaba_avg, ach_avg, glut_avg, oct_avg, ser_avg, da_avg) via direct download links + MD5. Route 2 (interactive/export): Codex (codex.flywire.ai) exposes directed edges carrying synapse count, neuropil, and predicted NT type; downloadable products connections_princeton and consolidated_cell_types. Route 3 (programmatic): CAVEclient (pip-installable) after a free Google-login CAVE account ('no extra permissions need to be given'); datastack flywire_fafb_public with immortal LTS versions 630 and 783; query_table/synapse_query with filter_in_dict on pre/post_pt_root_id extracts a bounded neuron-list subgraph. Hierarchical Schlegel et al. annotations (flow/super_class/cell_class/cell_type) enable principled subgraph selection.

Sources: zenodo.org/records/10676866 · codex.flywire.ai/faq · github.com/seung-lab/FlyConnectome (CAVE tutorial) · nature.com/articles/s41586-024-07558-y · caveclient.readthedocs.io

5. Q2 FLYWIRE GOTCHAS: The real engineering pitfalls are (1) NT signs are classifier predictions not ground truth, (2) weights require aggregation across multiple per-neuropil rows and a synapse-count threshold, (3) NT columns are probabilities requiring an argmax/threshold to derive excitatory/inhibitory signs, and (4) file sizes are large (~10GB) needing chunked processing.

confidence: HIGH | vote: 3-0 (merged from claims 10,17,19)

Evidence. NT identities are predictions from a CNN classifier (Eckstein et al. Cell 2024; 87% per-synapse, 94% per-neuron, 91% per-cell-type accuracy) discriminating six transmitters; in the fly 'acetylcholine is excitatory and GABA is inhibitory.' Signs are derived (acetylcholine excitatory; GABA/glutamate inhibitory; aminergic treated excitatory) as in Shiu et al. 2024 and Pospisil et al. effectome — not provided pre-signed. A pair counts as connected only above a dataset-specific synapse threshold (FAFB 5+) and 'connection tables can contain multiple rows for the same pair' across neuropils, so syn_count must be summed per pair before building a single weighted directed edge. Documented failure mode: truncated neurons can yield false all-glutamate predictions (e.g., OCG01).

Sources: nature.com/articles/s41586-024-07968-y · codex.flywire.ai/faq · nature.com/articles/s41586-024-07558-y

6. Q3 SCIENTIFIC VALIDITY (topology): Real fly connectome topology differs substantially from a synthetic Watts-Strogatz small-world graph and from random graphs, which both supports the topology-ablation hypothesis and raises a reviewer objection about what 'small-world' control actually tests.

confidence: HIGH | vote: 3-0 (claim 12, supported by 9,18)

Evidence. The fly brain is highly recurrent with small-worldness coefficient ~141 (vs C. elegans S=3.21), mean shortest directed path 4.42 hops (all neurons reachable within 13), and rich-club organization spanning ~30% (40,218) of neurons — features a Watts-Strogatz graph lacks by construction. Scale context: the full connectome is 139,255 neurons / ~54.5M synapses (127,978 neurons / 2.6M thresholded connections after a 5-synapse threshold), so a ~1000-neuron subgraph is <1% of the whole brain. This means the 'biological vs small-world vs random' ablation is meaningful (the three topologies are genuinely distinct), but reviewers will ask whether a sub-1% subgraph preserves the rich-club/recurrence that gives biological topology its purported advantage — a key validity risk.

Sources: nature.com/articles/s41586-024-07968-y · nature.com/articles/s41586-024-07558-y

7. Q3/Q4 SCIENTIFIC VALIDITY & PUBLISHABILITY (the empirical bar): Connectome-constrained modeling is already validated against biology at a top-tier bar, so a 1000-neuron student SNN will be measured against a high standard, and its publishability hinges on the honest AI-vs-RL-on-real-connectome contribution rather than on modeling fidelity.

confidence: MEDIUM | vote: 2-1 on the bar-setting claim; high on underlying facts

Evidence. Lappalainen et al. predicted real neural activity across 26 studies from connectivity + task optimization at single-neuron resolution (Nature 2024), and FlyGM showed real-connectome inductive bias on locomotion (2026) — both top-tier. A toy-grid SNN cannot compete on biological-prediction fidelity, so its contribution must be the methodological comparison (active inference vs RL across topology/coding/epistemic ablations with 50 trials + statistics). This is realistically a workshop / specialized-venue (active-inference, ALife, computational-neuroscience, or a NeurIPS/ICLR workshop track) contribution, not a top-tier main-conference or Nature-family paper. Clearing the bar requires: pre-registered ablations, proper statistical power, an honest prior-art section citing FlyGM/Lappalainen/NeuroMechFly/FEPS, and a clear statement that EFE is currently MSE-based (not full variational free energy) — which must be fixed for the active-inference claim to hold. NOTE: the venue/quality-bar assessment is the synthesizer's inference from the prior-art landscape, not a directly verified claim.

Sources: nature.com/articles/s41586-024-07939-3 · arxiv.org/abs/2602.17997 · ncbi.nlm.nih.gov/pmc/articles/PMC12410762

8. Q5 WETWARE STAGE 2: DishBrain-style closed-loop biological control is high-risk and largely infeasible for a student without a wet lab; the located evidence base in this research round is thin and the assessment relies on general knowledge of the field rather than verified claims.

confidence: LOW | vote: not directly verified in this claim set

Evidence. No claim in the verified set addresses Stage 2 wetware (Cortical Labs DishBrain, FinalSpark, or slime-mold-on-MEA) directly — this is a genuine gap in the evidence gathered. From domain knowledge (not verified here): DishBrain (Cortical Labs, 2022) required cultured human/mouse cortical neurons on a high-density CMOS MEA (e.g., MaxWell Biosystems), sterile cell-culture and incubation infrastructure, custom closed-loop electrophysiology software, and a multidisciplinary team — none trivially reproducible by an undergraduate without lab access. FinalSpark's Neuroplatform offers remote cloud access to living-neuron MEAs, which could lower the hardware barrier, but biological viability, electrode interfacing, and closed-loop latency remain hard. Slime-mold (Physarum) on MEA is cheaper and non-sterile-tolerant but is not a spiking system and does not map onto the SNN/active-inference framework. This question should be re-researched before any Stage 2 commitment.

Sources: arxiv.org/abs/2412.14112 · finalspark.com/neuroplatform · arxiv.org/abs/2405.16946 · en.wikipedia.org/wiki/Cortical_Labs

9. Q6 VERDICT: Continuing is worth it IF reframed as a Stage 1.5 software study scoped tightly around 'active inference vs RL on a real FlyWire connectome'; jumping to wetware (Stage 2) gives poor return for a student's time and should be deferred or dropped.

confidence: MEDIUM | vote: synthesized from full claim set

Evidence. The defensible contribution survives prior art (claims 1,6,7,8): no located work compares active inference/EFE to RL on a real connectome. The data is accessible and the ablation design is meaningful (claims 11-23, 12). The main risks: (1) the current EFE-from-MSE is NOT full variational free energy, so the headline 'active inference' claim is not yet earned — fixing this is the single highest-priority engineering task; (2) a <1% subgraph may not preserve the rich-club/recurrence that motivates the biological-topology hypothesis (claim 12), risking a null result; (3) FlyGM already showed connectome inductive bias, so the topology ablation alone is not novel — the active-inference axis carries the novelty. Rough effort: Stage 1.5 software paper ~3-6 months part-time (data extraction days; full VFE/EFE + RL baseline + ablations weeks; 50-trial stats + writeup weeks). Stage 2 wetware: 12+ months and external lab/funding — not advisable as the next step. Best return: publish the Stage 1.5 software study at a workshop, treat wetware as a separately-funded future direction.

Sources: arxiv.org/abs/2602.17997 · ncbi.nlm.nih.gov/pmc/articles/PMC12410762 · nature.com/articles/s41586-024-07939-3

Killed Claim (failed adversarial verification: 1 of 25)

REFUTED (vote 1-2): "NeuroMechFly v2 trains its embodied Drosophila controller for closed-loop multimodal navigation using reinforcement learning, not active inference or free-energy methods — establishing RL (not active inference) as the standard control paradigm for connectome-era fly models."

Why killed: the second clause (a sweeping 'establishes the standard paradigm' generalization) over-reached beyond what one paper supports, so verifiers refused it — even though the narrow fact that NeuroMechFly v2 uses RL is independently confirmed in Finding 2. Source: biorxiv.org/content/10.1101/2023.09.18.556649.

Caveats & Confidence Limits

- **Time-sensitivity:** all connectome-data findings reflect the frozen FlyWire v630/v783 snapshots (2023) accessed in 2026; these are immortal LTS versions (stable) but Codex/CAVEclient APIs evolve.
- **FlyGM is a 2026 preprint** (arXiv:2602.17997) that may not yet be peer-reviewed; its existence nonetheless materially constrains NeuroSpark's novelty framing.
- **Question 5 (wetware) is weakly evidenced** — no verified claim supports it; low-confidence, based on general domain knowledge. Re-research before any Stage 2 decision.
- **Venue/quality-bar judgments (Q4)** are synthesizer inferences from the prior-art landscape, not directly verified claims. One claim (the 'high empirical bar' framing) carried a split 2-1 vote.
- **Project correctness gap:** the current Stage 1 EFE is computed from MSE prediction error, NOT full variational/expected free energy — until fixed, the 'active inference' novelty is not technically earned and reviewers would flag it.
- **Neurotransmitter signs** throughout are classifier predictions (~87-94% accuracy), not ground truth.

Open Questions (not resolved by this round)

1. What did Cortical Labs DishBrain and FinalSpark actually require in hardware, cost, biological expertise, and team size, and is FinalSpark's remote Neuroplatform a realistic access route for a student? (Q5 was not answered by the verified claim set.)
2. Does a ~1000-neuron FlyWire subgraph (<1% of the brain) actually preserve the rich-club and recurrent motifs that motivate the biological-topology advantage, or does subsampling destroy exactly the features being tested? Central scientific-validity risk; currently untested.
3. Is there any 2026 work — beyond FlyGM, NeuroMechFly v2, FEPS, and Lappalainen — that already combines active inference/free energy with a real connectome, which would fully pre-empt the narrow novelty niche?
4. What is the principled way to select a functionally coherent ~1000-neuron subgraph (e.g., a specific sensorimotor pathway via Schlegel et al. cell-type annotations) rather than an arbitrary cut, so the topology ablation is biologically meaningful?

Full Source List (24 fetched, by angle)

#	Angle	Qual.	Source
1	novelty / prior art	primary	arxiv.org/abs/2602.17997 — FlyGM
2	novelty / prior art	primary	nature.com/articles/s41586-024-07939-3 — Lappalainen et al.
3	novelty / prior art	primary	biorxiv.org/content/10.1101/2023.09.18.556649 — NeuroMechFly v2
4	novelty / prior art	primary	ncbi.nlm.nih.gov/pmc/articles/PMC12410762 — FEPS
5	FlyWire data	primary	nature.com/articles/s41586-024-07968-y — NT predictions
6	FlyWire data	primary	github.com/seung-lab/FlyConnectome — CAVE tutorial
7	FlyWire data	primary	codex.flywire.ai/faq — Codex
8	FlyWire data	primary	nature.com/articles/s41586-024-07558-y — FlyWire connectome
9	FlyWire data	primary	zenodo.org/records/10676866 — v783 bulk download
10	FlyWire data	primary	caveclient.readthedocs.io — materialization docs
11	scientific validity	primary	arxiv.org/abs/2604.04033
12	scientific validity	primary	sciencedirect.com/science/article/pii/S1877050921012515
13	scientific validity	primary	ncbi.nlm.nih.gov/pmc/articles/PMC6060791
14	scientific validity	primary	arxiv.org/pdf/1909.10863 — Millidge
15	publishability	primary	eneuro.org/content/overview
16	publishability	primary	journals.plos.org/ploscompbiol — submission guidelines
17	publishability	primary	iwaiworkshop.github.io — IWAI
18	publishability	primary	neuroai-workshop.github.io
19	publishability	primary	journals.plos.org/ploscompbiol/article (pcbi.1007881)
20	publishability	primary	eneuro.org/content/9/4/ENEURO.0017-22.2022
21	wetware	primary	arxiv.org/abs/2412.14112
22	wetware	primary	finalspark.com/neuroplatform
23	wetware	primary	arxiv.org/abs/2405.16946
24	wetware	secondary	en.wikipedia.org/wiki/Cortical_Labs

Run statistics

Angles searched: 5 · sources fetched: 24 · claims extracted: 116 · claims verified: 25 · confirmed: 24 · killed: 1 · findings after synthesis: 9 · total agent calls: 106 · subagent tokens: ~2.63M · duration: ~81 min.

This document is the verbatim verified-evidence record from the deep-research workflow run on 22 June 2026; it is the proof backing the verdict in 'NeuroSpark_Stage1_5_Viability_and_GoForward_Plan.pdf'.